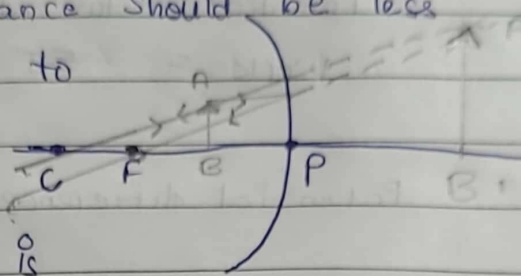


10. Given

$$f = 20 \text{ cm}$$

⇒ The object distance should be less than focal length to form erect image



The image formed is
Enlarged & virtual (Behind the mirror)

11. Appearance of ^{red colour of} Sun in early morning

- The light rays reaching the observer has to ~~reach~~ Travel a long distance through the atmosphere. Thus, most of the blue light gets scattered leaving only red light to reach the observer's eyes.
- The ability of the human eye to adjust according to its focal length is called accommodation of the eye.
- The least distance of distinct vision for human eye is ~~infinity~~ ^{25 cm}. The near point → 25 cm without eye strain.

9. Given

Potential at two ends = 10V & 5V

Work = 5J

$$\text{Potential difference (V)} = 10\text{V} - 5\text{V} \\ = 5\text{V}$$

$$V = \frac{W}{Q} \rightarrow Q = \frac{W}{V}$$

$$Q = \frac{5}{5} = 1$$

$$Q = 1\text{C}$$

$\therefore$ 1C of charge flowing between the two ends.

10. (a) Pole - The centre point of the spherical mirror

(b) Radius of curvature - The distance between the pole to centre of curvature.

(c) Principal axis - An imaginary line passing through centre of curvature and pole of a spherical mirror.

(d) Principal focus - The point where the light rays meet

$\therefore$ for concave mirror - 'C', 'P', 'F', 'R' are in front of the mirror and ^{taken as} are and are negative.

8. Given

$$F = +12 \text{ cm}$$

$$h = 5 \text{ cm}$$

$$u = -20 \text{ cm}$$

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} = \frac{1}{12} - \frac{1}{20}$$

$$\frac{1}{v} = \frac{5-3}{60} \Rightarrow \frac{2}{60} = \frac{1}{30}$$

$$v = 30 \text{ cm}$$

~~(Real & inverted)~~

$$m = \frac{v}{u}$$

$$m = \frac{30}{-20} = -1.5$$

~~(Virtual & erect)~~

$$m = \frac{h_i}{h}$$

$$-1.5 = \frac{h_i}{5}$$

$$-1.5 \times 5 = h_i$$

$$-7.5 = h_i$$

Nature - ~~virtual & erect~~ Real & inverted

Size of image (h_i) = -7.5

Position of image = 30 cm behind the mirror

Board - Exam - PHY - CHE - BID

1. Presbyopia
2. Repulsion
3. virtual and erect, Same Distance as object, Same Size as object & laterally inverted.
4. It has to be shot below the apparent position of the fish image
5. Given

$$P = +1.5D$$

$$f = \frac{1}{P} \Rightarrow \frac{1 \times 10}{1.5 \times 10} = \frac{10^{-2}}{1.5} \Rightarrow +0.67 \text{ (approx)}$$

$\Rightarrow$ The lens prescribed is converging lens.

6. Given

$$I = 5A$$

$$\text{Time} = 30 \text{ min}$$

$$1 \text{ min} = 60 \text{ sec}$$

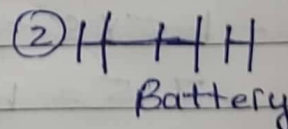
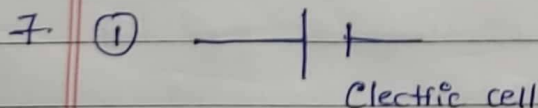
$$\hookrightarrow 30 \times 60 = 1800 \text{ sec}$$

$$I = \frac{Q}{t} \rightarrow Q = It$$

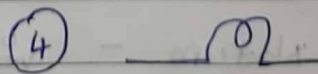
$$5 \times 1800 = Q$$

$$Q = \cancel{9000} \quad 9,000C$$

$$\text{or } Q = 9 \times 10^3 C$$



Resistor of
constant resistance



Bulb

